

Dipto Kumar Haldar

Batiaghata-9260, Khulna, Bangladesh

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Education

2025 **Khulna University of Engineering and Technology (KUET)**
Bachelor of Science in Industrial and Production Engineering
CGPA: 3.35 / 4.00
Key Undergraduate Courses: Operations Management, Operations Research, Quality Management, Logistics and Supply Chain Management, Project Management

Research Interests

Supply chain optimization; operations research; multi-hazard risk assessment; mathematical modeling; data-driven decision making; optimization under uncertainty; machine learning; community resilience decision support.

Research Experience

2026-Present **Research Affiliate**
Risk-Informed Systems Engineering Lab (RISELAB)
Ongoing Research Project: Equitable Disaster Resource Allocation under Causal Ambiguity
Developed a novel robust optimization framework combining causal inference, partial identification, and predictive analytics to improve disaster resource allocation under uncertainty. The project addressed hidden confounding in evacuation behavior and introduced a fairness-driven allocation strategy under incomplete information. Applied the framework to Hurricane Ian evacuation data across 223 census tracts in Florida, quantifying trade-offs between efficiency, equity, and robustness. The work contributes to resilient emergency planning by advancing counterfactual policy evaluation and identification-aware decision-making.
Supervisor: Himadri Sen Gupta, Ph.D.
Assistant Professor, Colorado State University Pueblo

2025 **Undergraduate Thesis**
Khulna University of Engineering and Technology (KUET)
Thesis Supervisor: Palash Saha

Research and Publications

Published Journal Articles

- 2026** • **Dipto Kumar Haldar***, Palash Saha, H. M. Belal, Oliver G Kayas. An Analytical Study on Integrating Artificial Intelligence and Industry 4.0 Technologies for Supply Chain Resilience and Sustainability. *Supply Chain Analytics*, 2026. <https://doi.org/10.1016/j.sca.2026.100209>

Journal Manuscripts Under Review

- 2026** • **Dipto Kumar Haldar***, Himadri Sen Gupta. Evaluating Behavioral-Risk Models for Hurricane Evacuation Allocation. *Risk Analysis*, 2026. **Submission ID: 3373113**
- 2026** • **Dipto Kumar Haldar***, Himadri Sen Gupta, Lava Kumar Polu. Equitable Evacuation Support via Conformal-Robust Optimization. *Transportation Research Part E*, 2026. **Submission ID: TRE-D-26-02168**

Skills

- **Programming:** Python (NumPy, Pandas, Scikit-learn), Basic C
- **Optimization Solver:** Gurobi, Pyomo
- **Design Software:** AutoCAD, SolidWorks, Fusion 360
- **Data Analysis Software:** SmartPLS, IBM SPSS, Microsoft Office, Excel, Power BI
- **Soft Skills:** Analytical thinking, problem solving, teamwork, communication, leadership, time management

Linguistic Proficiency

- **English:** Fluent Working Proficiency
- **Bengali:** Native Language

Academic Project

2023 **Undergraduate Project**
Khulna University of Engineering and Technology (KUET)
Project Title: “Developed an advanced fire detection and automated notification system using Arduino, GSM Module, and IR sensors”

Industrial Experience

2023 **Industrial Training – BSRM**
Completed industrial training in the Production Department at BSRM. Learned about steel production processes and professional-level industrial operations.
Key Learnings: Safety and organizational goals, furnace and LRF operation, CCM process, spectro lab inspection, maintenance, and utility management.

Extra-Curricular Activities

2024 **Aspire Institute**
Aspire Leadership Program
Developed skills in communication, leadership, and international networking.

2023-2024 **IEM Debating Club (IDC), KUET**
Vice-President
Organized “Multiverse Open 2024.” Strengthened skills in public speaking, critical thinking, and event management.

Achievements & Awards

- **Finalist:** Walton Laptop Presents MindSparks 2022 (Business Case Competition), AUST Innovation and Design Club
- **Semi-finalist:** Hult Prize at KUET 2024 (On-Campus)
- **Semifinalist:** Face the Case Competition (FTC 3.0), IEEE Student Branch CUET, 2023
- **Technical Scholarship:** Awarded based on academic CGPA at KUET
- **Government Scholarship:(SSC Level)** Awarded by the Board of Intermediate and Secondary Education, Jashore, Bangladesh, for outstanding academic performance